CEBU INSTITUTE OF TECHNOLOGY UNIVERSITY

COLLEGE OF COMPUTER STUDIES

Software Project Management Plan

for

DASIGConnect

DOCUMENT CONTROL

Document Status

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Document title	Software Project Management Plan for DASIGConnect
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Prepared by	DASIGConnect Project Team
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Revision History

Version	Date	Description	Authors
1.0	17 Sep 2026	Initial complete SPMP aligned with the refactored requirements.	DASIGConnect Project Team
1.1	10 Oct 2026	Updated project organization, budget allocation, service providers, deployment considerations, and management documentation.	DASIGConnect Project Team

Project Team

Name	Project participation
Richemmae V. Bigno	Development team member; shared analysis, implementation, testing, documentation, and review responsibilities.
Jay Lord C. Bayonas	Development team member; shared analysis, implementation, testing, documentation, and review responsibilities.
Chris Daniel P. Cabatana	Development team member; shared analysis, implementation, testing, documentation, and review responsibilities.
Mark Anton L. Camoro	Development team member; shared analysis, implementation, testing, documentation, and review responsibilities.
Lerah A. Caones	Development team member; shared analysis, implementation, testing, documentation, and review responsibilities.

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1. OVERVIEW

1.1 Project Summary

DASIGConnect is a web-based social media content workflow and scheduling management system for the DOST Academe-Science and Innovation Group (DASIG) under DOST Region 7. It coordinates content from multiple member institutions and replaces fragmented manual communication with a traceable workflow for submission, moderation, scheduling, publishing, media management, notifications, analytics, and AI-assisted content preparation.

This SPMP is the first complete project management plan for DASIGConnect. It uses the supplied IEEE 1058-style template and incorporates verified requirements and implementation changes from the 19-29 May 2026 Module 3 baseline through 4 October 2026. It complements the proposal, SRS, SDD, implementation plans, test evidence, validation deliverables, and Git development-history audit.

1.1.1 Purpose, Scope, and Objectives

The project purpose is to improve the timeliness, completeness, governance, and measurability of DASIG Facebook content operations while preserving human editorial authority. The current implementation recognizes three roles: Contributor, Moderator, and Admin. Contributors prepare content; Moderators perform network-wide editorial review; Admins govern accounts, institutions, configuration, exceptions, and operational oversight.

In-scope capabilities are authentication and invitation; institution and account management; content drafting and autosave; review and revision; scheduling guard rails and calendar views; Facebook Page connection and publishing; failed-publication resolution; media albums, search, reuse, lifecycle, and watermarking; notifications and Messenger connection; analytics and engagement synchronization; AI captioning, classification, embeddings, and recommendations; audit logs; system health; security; data retention; and role-aware performance caching.

The plan retains the measurable objectives documented in the proposal and SRS.

- Support at least three DASIG member institutions with role-based access and institution-level data separation.
- Achieve at least 95% complete submitted records through server-enforced fields and actionable validation.
- Require an explicit Moderator or Admin approval before publication.
- Target email status delivery within five minutes in at least 95% of cases and near-real-time in-app status updates where specified.
- Return media searches within two seconds for a library of up to 1,000 assets.
- Target at least 95% successful scheduled publication within plus/minus five minutes over at least 20 pilot publications.
- Evaluate AI caption availability, use, quality, image classification accuracy, and media recommendation relevance against the approved proposal metrics.
- Evaluate system usability through task completion measures and the System Usability Scale (SUS), targeting a minimum SUS score of 70 out of 100 to demonstrate acceptable usability and user satisfaction.

1.1.2 Exclusions, Assumptions, and Constraints

The initial product is scoped to Facebook and does not provide a general multi-platform publisher. AI does not autonomously approve or publish content and does not generate images. Hardware procurement, institutional internet infrastructure, Meta policy decisions, and long-term production operations are outside the student team's direct control.

- Public publishing depends on DASIG Page authorization, valid Meta permissions and tokens, suitable app mode/review, and Meta platform policy.
- The system depends on internet access and external hosting, database, storage, email, Facebook, Gemini, and Voyage AI services.
- The available academic pilot window may not reveal long-term posting patterns.
- Historical Facebook observations may not capture the private coordination that produced each post.
- AI services must degrade without blocking the manual workflow.
- Exact post-September defense, production transition, and turnover dates remain subject to the academic calendar and stakeholder approval.

1.1.3 Project Deliverables

Deliverable	Due date	Required evidence	Current status
MVP Validation package	12 Sep 2026, 11:59 PM	Google Form; framework/model PDF; linked response sheet; highlights PDF; organized Google Drive evidence folder.	COMPLETED - all five outputs
Updated SRS and RTM	19 Sep 2026, 11:59 PM	Refactored SRS based on validation findings with the required Requirements Traceability Matrix incorporated.	COMPLETED
Updated SDD	19 Sep 2026, 11:59 PM	Design aligned with the refactored SRS, including affected architecture, data, interfaces, and workflows.	COMPLETED
Updated SPMP	19 Sep 2026, 11:59 PM	Updated scope, schedule, WBS, deliverables, resources, roles, risks, and other affected management plans.	COMPLETED
Refactoring Priorities	10 Oct 2026	Evidence-based priorities connecting validation findings, SMART objectives, requirement/design changes, and expected improvement.	IN PROGRESS
Software Test Documents	10 Oct 2026, Week 9	Test Plan, Test Cases, and Test Reports.	IN PROGRESS
Deployed System	17 Oct 2026, Week 10	Pilot or production deployment with test execution and usability test results.	IN PROGRESS
Source code and migrations	Continuous	Reviewed pull requests, reproducible builds, migration chain, test evidence, and identified baselines.	GitHub development baseline

1.1.3.1 Completed MVP Validation Outputs

The team confirms that the following required MVP Validation outputs are complete. These records must remain available as the source evidence for refactoring and documentation updates.

- Google Form validation instrument aligned with the selected framework, SMART objectives, measurable outcomes, respondent roles, and qualitative feedback needs.
- Validation Framework or Model PDF explaining the framework, its suitability, objective-to-construct mapping, questionnaire groups, and respondent roles.
- Google Sheet linked to the form, with organized responses and identifiable respondent roles where appropriate.
- MVP Validation Highlights PDF summarizing meaningful findings, issues, recommendations, positive aspects, and proposed improvements.
- Organized Google Drive evidence folder containing appropriate proof of deployment, participation, recordings, interviews, consultations, or communications.

1.1.4 Schedule and Budget Summary

Schedule Summary

Period/Milestone	Primary output	Basis
Early 2026	Problem definition, proposal, objectives, and initial requirements.	Proposal and early repository history
May 2026	SRS/SDD baseline and Module 3 implementation baseline.	SRS/SDD dated 16 May; commit ce776d2 dated 29 May
June-August 2026	Requirements refactoring, workflow expansion, platform hardening, and integration work.	Git history and migrations
August-September 2026	Use-case refactor, performance caching, security, account governance, media lifecycle, monitoring, regression testing, and defect correction.	Merged PRs 74-332
12 Sep 2026	MVP Validation package completed.	Five instructor-required outputs
19 Sep 2026	Updated SRS/RTM, SDD, and SPMP milestone; document are completed	Weeks 1-6 deliverables
10 Oct 2026	Refactoring Priorities and Software Test Documents due.	Week 9 instructor milestone
17 Oct 2026	Deployed System with test execution and usability results due.	Week 10 instructor milestone
Post-deployment	Final revisions, production authorization, acceptance, transition, and turnover.	TBD - stakeholder and academic approval

Budget Summary

Item / Service	Provider	Billing Cycle	Cost (Php)
Domain Registration	Namecheap	Yearly	₱440.60
Backend Server	Railway Hobby	2 Months	₱624.60
Database	Supabase Pro	2 Months	₱3,123.00
API usage	Gemini and Voyage	2 Months	₱1,300.00
Total Projected Infrastructure Cost		2 Months + Domain	₱5,488.20
Total Project Budget Allocation		Entire Project	₱10,000.00
Actual Expenses to Date		As of October 10, 2026	₱1,740.60
Remaining Unspent Budget			₱8,259.40

1.2 Evolution of the Plan

This SPMP is controlled in Git with the project documentation. Material changes to scope, schedule, roles, risks, external integrations, acceptance criteria, or resource commitments require a documented change, team review, and an updated version entry. Urgent updates may be made between regular reviews but must be reconciled at the next team review.

Trigger	Required action	Approval/record
Approved requirement change	Assess scope, schedule, risk, tests, and affected documents.	Issue/PR and updated traceability
Third-party policy or API change	Review permissions, data handling, fallback, and delivery impact.	Technical review and risk update
Milestone variance	Re-estimate remaining work and agree corrective action.	Sprint review/status record
Security or privacy issue	Triage immediately, contain exposure, test correction, and record lessons.	Incident/defect record and reviewer sign-off
Final acceptance feedback	Prioritize required changes and baseline the accepted release.	Acceptance record and release tag

2. REFERENCES

Reference	Version/Date	Use in this SPMP
DASIGConnect Project Proposal	2026	Purpose, objectives, methodology, MVP, scope, limits, and evaluation measures.
DASIGConnect Software Requirements Specification	Version 6.0, 27 September 2026	Functional/non-functional requirements, business rules, security, reliability, and acceptance thresholds.
DASIGConnect Software Design Description	Version 3.0, 10 October 2026	Architecture, modules, data, APIs, security mechanisms, and external integrations.
Module 3 Git baseline	ce776d2, 29 May 2026	Pre-refactor implementation baseline.
Refactored dev baseline	3976afb, 4 Oct 2026	Current implementation comparison baseline at the Git audit date.
DASIGConnect Git Development History Audit	4 Oct 2026	Verified periods, use-case ownership, PR evidence, and integration milestones.
IT411 Weeks 1-6 Deliverables and Week 9 Notice	Sep-Oct 2026	Required validation, SRS, SDD, SPMP, refactoring, testing, and deployment outputs and deadlines.

3. DEFINITIONS

Term	Definition
Admin	Current system role responsible for network governance, privileged configuration, account/institution management, and exception handling.
Moderator	Current network-wide editorial role. This replaces the legacy Validator terminology used by the May baseline.
Contributor	User who creates, edits, schedules, and tracks content submissions.
DASIG	DOST Academe-Science and Innovation Group.
MVP	Minimum Viable Product used for structured stakeholder validation.
RLS	Row-Level Security used as a database defense for tenant/institution separation.
SUS	System Usability Scale.
UAT	User Acceptance Testing.
RTM	Requirements Traceability Matrix.
STD	Software Test Documents consisting of the Test Plan, Test Cases, and Test Reports.
SPMP / SRS / SDD	Software Project Management Plan / Software Requirements Specification / Software Design Description.
Guard rail	Configured scheduling or workflow rule that blocks, warns, or governs an action.
Baseline	An identified version used for controlled comparison and acceptance.

4. PROJECT ORGANIZATION

4.1 External Structure

The development team operates within Cebu Institute of Technology - University, College of Computer Studies. DASIG/DOST Region 7 and participating institutions supply operational context, user feedback, Page authorization, and acceptance input. External service providers supply hosting, database, object storage, email, AI, and Meta platform capabilities under their own policies and service limits.

External entity	Interface with the project
DASIG / DOST Region 7	Operational owner and principal stakeholder; confirms workflow, Page access, validation feedback, and transition decisions.
Member institutions	Supply Contributors and pilot users; provide representative content and workflow feedback.
CIT-U CCS / instructor / adviser	Academic governance, review, milestone evaluation, and acceptance guidance.
Meta / Facebook	OAuth, Page permissions, publishing, engagement data, app review, and platform policy.
Vercel, Railway, Supabase, Cloudflare	Frontend hosting, backend hosting, PostgreSQL service, and object storage.
Gemini, Voyage AI, Resend	AI assistance, embeddings, and email delivery.

4.2 Internal Structure

The DASIGConnect development team consists of five members with designated responsibilities in project management, frontend development, backend development, Scrum coordination, and quality assurance. Richemmae V. Bigno serves as the Team Lead and Project Manager, overseeing project planning, task assignments, development progress, and milestone completion. Jay Lord C. Bayonas and Chris Daniel P. Cabatañа serve as Frontend and Backend Developers, respectively, while jointly fulfilling Scrum Master responsibilities to coordinate sprint activities and monitor development progress. Mark Anton L. Camoro and Lerah A. Caones serve as Frontend and Backend Developers, respectively, while sharing Quality Assurance responsibilities, including software testing, defect identification, and verification.

Although each member has designated primary roles, responsibilities may be temporarily rotated or shared depending on individual availability, workload, and project priorities to maintain development continuity and meet deadlines. All team members collaborate through Git-based version control, peer reviews, documentation, and iterative Agile development to ensure project quality and timely delivery.

4.3 Roles and Responsibilities

Role	Responsibilities	Accountability
Project Manager / Team Lead	Oversees project planning, development progress, task assignments, technical decisions, and milestone completion.	Overall project delivery and technical direction
Scrum Master	Facilitates sprint planning, monitors task progress, resolves blockers, and coordinates development activities.	Sprint coordination and Agile process compliance
Frontend Developer	Develops user interfaces, implements frontend functionality, integrates backend APIs, and ensures responsive design.	Frontend functionality and usability
Backend Developer	Develops APIs, database operations, business logic, authentication, security, and external service integrations.	Backend functionality and system reliability
Quality Assurance (QA)	Prepares and executes test cases, identifies defects, verifies fixes, and documents testing results.	Software quality and testing compliance
DASIG Stakeholder Representative	Validates operational requirements, provides feedback, and coordinates Facebook Page authorization.	Operational acceptance
Instructor / Adviser	Evaluates project deliverables, documentation, academic compliance, and milestone completion.	Academic evaluation
Pilot Participants	Evaluate system functionality and usability and provide structured feedback.	User validation evidence

Names	Roles
Richemmae Bigno	Team Lead / Project Manager
Jay Lord Bayonas	Frontend Developer / Scrum Master

Chris Daniel Cabataña	Backend Developer / Scrum Master
Mark Anton Camoro	Frontend Developer / Quality Assurance
Lerah Caones	Backend Developer / Quality Assurance

5. MANAGERIAL PROCESS PLANS

5.1 Start-up Plan

5.1.1 Estimation Plan

Work is decomposed into user-visible outcomes, backend/API changes, data migrations, frontend behavior, tests, documentation, and deployment tasks. Estimates are reviewed at sprint start and updated when implementation evidence reveals new dependencies. The team uses relative complexity, historical pull-request size, dependency risk, and academic deadlines rather than treating early estimates as fixed commitments.

5.1.2 Staffing Plan

All five team members remain part-time student developers. Work is assigned by competency and available capacity while maintaining peer review and knowledge sharing. A work item should have a primary implementer and at least one reviewer; high-risk security, migration, publishing, or tenant-isolation work requires explicit test evidence.

5.1.3 Resource Acquisition Plan

Resource	Acquisition/control	Contingency
Development tools and GitHub	Use existing team accounts and repository access.	Local development and repository archive.
Vercel / Railway/ Supabase / R2	Use approved project environments and secret-managed configuration.	Documented redeployment and provider export/backup where available.
Meta Page and app access	DASIG Page administrator authorizes the app/account and required permissions.	Development/test Page and manual publishing fallback.
Gemini / Voyage AI / Resend	Enable only with approved keys, limits, and budget controls.	Manual captioning/tagging and in-app workflow remain usable.
Pilot participants	Recruit role-appropriate users and record consent/evidence.	Supplement with scheduled follow-up interviews where needed.

5.1.4 Project Staff Training Plan

- Review SRS/SDD, current role terminology, state transitions, and tenant boundaries before changing workflows.
- Use secure secret handling, migration review, least privilege, and safe test data practices.
- Review Meta permissions, token lifecycle, app mode, publishing constraints, and fallback procedures.
- Review React Query cache keys/invalidation and avoid cross-session or cross-role stale data.
- Practice build, automated-test, deployment, rollback, and evidence-capture procedures before acceptance.

5.2 Work Plan

5.2.1 Work Activities and WBS

WBS	Work package	Principal outputs
1.0	Project governance	SPMP, schedule, status, risk, change and acceptance records.
2.0	Validation and refactoring	Completed validation package, findings, evidence-based priorities, and objective alignment.
3.0	Requirements and design	Updated SRS, RTM, SDD, traceability, and architecture decisions.
4.0	Identity and institutions	Authentication, invitations, account states, roles, institutions, tenant controls.
5.0	Content workflow	Drafting, autosave, submission, moderation, revision, scheduling, templates.
6.0	Publishing and communication	Facebook connection, scheduler, retries, fallback, notifications, Messenger.
7.0	Media and AI	R2 storage, library, albums, lifecycle, watermarking, captioning, classification, recommendations.

WBS	Work package	Principal outputs
8.0	Analytics and operations	Workflow/engagement analytics, audit logs, system health, background jobs.
9.0	Quality and security	STD, automated/manual tests, security checks, performance/caching, accessibility, defects.
10.0	Deployment and transition	Pilot deployment, test/usability results, guidance, handover, and archive.

5.2.2 Schedule Allocation

Milestone	Exit condition	Timing
Requirements/design baseline	Proposal, SRS, and SDD available for implementation.	May 2026
Module 3 baseline	Three-module implementation merged.	29 May 2026
Refactor/hardening increments	Approved PRs merged with tests and updated behavior.	June-September 2026
MVP validation package	All five required outputs complete and retained as refactoring evidence.	Completed - 12 September 2026 milestone
Updated SRS, SDD, and SPMP	Validation findings traced into updated requirements, design, and management plans.	Completed - 19 September milestone
Refactoring Priorities and STD	Evidence-based priorities plus Test Plan, Test Cases, and Test Reports.	10 October 2026
Deployed System	System ready for testing with execution and usability results.	17 October 2026
Final acceptance and transition	Critical defects closed; evidence accepted; handover material complete.	TBD - academic calendar

5.2.2.1 Verified Development Phase Schedule

Period	Verified activity	Repository evidence
19-29 May 2026	Repository foundation and the original 12-use-case Module 3 MVP implementation.	PRs 1-72; commit ce776d2
30 May-8 Jun 2026	Advanced media-library/Facebook-insight experiments and UX hardening.	capstone2-advance-development history
21-30 Aug 2026	Current 21-use-case refactor, role/account redesign, workflow expansion, R2 migration, analytics, system health, and audit log.	Principal PRs 74-147
31 Aug-12 Sep 2026	Security hardening, account and moderator corrections, caching architecture, responsive behavior, and requirements reconciliation.	PRs 145-212
13-24 Sep 2026	Performance, pagination, cancellation, progressive loading, media lifecycle, review hardening, and analytics improvements.	PRs 213-332
24 Sep-1 Oct 2026	Multi-image media AI, durable processing, Voyage ingestion/search/ranking, evaluation, monitoring, and metadata integration.	PRs 333-373
4 Oct 2026	Admin/moderator transfer confirmation correction.	PR 374

Main integration milestones were Module 3 to main on 29 May; Dev PRs 152 and 154 on 1 September; v2.1 PR 224 on 13 September; v2.2 PR 257 on 15 September; and v2.3.2 PR 305 on 21 September 2026.

5.2.3 Resource Allocation

Team effort is prioritized in this order: critical security/data-integrity defects; blocking workflow defects; acceptance-critical features; test and evidence completion; performance and maintainability improvements; optional enhancements. External-service effort includes configuration, permission review, integration testing, monitoring, and fallback verification.

5.2.4 Budget and Expense Allocation

Expense category	Planned amount	Actual amount
Hosting / database / storage	₱ 4,000.00	₱ 0.00
AI and external API usage	₱ 1,500.00	₱ 0.00
Domain / communication / email	₱ 500.00	₱ 440.60
Printing and documentation	₱ 500.00	₱ 0.00
Transportation / meetings	₱ 1,000.00	₱ 500.00
MVP respondent tokens / snacks	₱ 1,000.00	₱ 800.00
Contingency / other	₱ 1,500.00	₱ 0.00
TOTAL	₱10,000.00	₱1,740.60

5.3 Control Plan

5.3.1 Requirements Control

The SRS is the requirements source of record, subject to approved change. Each change is linked to an issue or task, reviewed for role/tenant/state-transition impact, implemented on a branch, tested, and merged through a pull request. Current implementation terminology prevails where the approved refactor intentionally supersedes legacy names, and the affected documentation must be aligned.

5.3.1.1 Validation-Driven Refactoring Control

Major requirements and design refactoring must be justified by MVP validation evidence and its effect on a SMART objective or measurable outcome. The controlled trace is: MVP validation evidence; stakeholder finding; improvement opportunity; affected objective; updated SRS requirement; updated SDD element; implementation change; test case and result; acceptance decision.

Accuracy, performance, efficiency, effectiveness, reliability, and outcome quality are valid refactoring targets when supported by evidence. Bug fixes, cosmetic changes, minor UI adjustments, correction of implementation defects, and completion of an existing requirement are not automatically treated as major refactoring. A new feature qualifies only when evidence shows it is necessary to achieve an approved objective.

Required question	Required record
What did stakeholders reveal?	Finding plus response, recording, observation, interview, or consultation evidence.
Which objective is affected?	Exact SMART objective, metric, and current/target outcome.
What must change?	Affected SRS requirement and SDD architecture, component, data, interface, or workflow.
How should the outcome improve?	Expected measurable effect, test method, test case, and acceptance condition.

5.3.2 Schedule Control

Progress is reviewed per sprint and before academic milestones. Variance is assessed by incomplete acceptance conditions, unresolved blockers, and external dependencies. Corrective actions include reducing optional scope, parallelizing independent work, reassigning reviewers, or invoking a documented fallback; security and data-integrity controls are not removed to recover time.

5.3.3 Budget Control

Only recorded, approved project expenses are entered. Paid APIs must use provider limits where available. Cost-affecting scope changes require team review and an updated expense record. No secret, billing credential, or full token is stored in project documents.

5.3.4 Quality Control

A work item is complete when its acceptance conditions are met, relevant automated/manual tests pass, authorization and tenant scope are verified, errors and loading states are handled, documentation is updated, and the change is reviewed. Database migrations must be forward-safe and tested against representative data.

5.3.5 Reporting Plan

Report/event	Frequency	Audience	Content
Sprint status	At least once per sprint	Project team	Completed work, blockers, decisions, next priorities.
Pull request	Per change set	Implementer and reviewers	Scope, files, tests, migration/security impact, manual checks.
Validation/refactoring review	Per proposed priority	Team, adviser/instructor as applicable	Evidence, objective, requirement/design impact, expected metric, test mapping.
Milestone review	At academic milestones	Team, instructor/adviser, stakeholders as applicable	Deliverables, evidence, variance, risk, acceptance.
Incident/critical defect	Immediately	Team and affected reviewer/stakeholder	Impact, containment, correction, verification, follow-up.

5.3.6 Metrics Collection Plan

Metric	Source	Target/use
Submission completeness	Submission records at pending transition	At least 95% complete.
Approval coverage	Validation/audit history versus published records	100% approved before publication.

Metric	Source	Target/use
Notification latency	Event and delivery logs	Email within 5 minutes in at least 95% of cases; specified in-app latency.
Publish timeliness/success	Publication attempts and scheduled time	At least 95% within plus/minus 5 minutes over pilot set.
Media search latency	Measured search requests	At most 2 seconds up to 1,000 assets.
AI caption use/quality	AI interaction log and questionnaire	At least 60% use/use-with-edits; average at least 4.0/5.0.
Classification/recommendation	Held-out labels and role questionnaire	At least 80% accuracy; at least 70% top-result relevance.
Usability	SUS responses by role	Target a minimum System Usability Scale (SUS) score of 70 out of 100 to demonstrate acceptable system usability.
Defects and performance	Test runs, issue log, browser/API observations	No open critical defect at release; regression trends reviewed.

5.3.7 Risk Management Plan

Risk	L/I	Mitigation and contingency	Owner
Meta authorization, app review, token, permission, or policy delay	H/H	Use a test Page in development; track required permissions; validate tokens; retain manual publishing fallback; treat production activation as stakeholder-dependent.	Integration owner / DASIG Page admin
Tenant or authorization defect	M/H	Defense in depth, server-derived scope, endpoint tests, RLS where configured, audit access denials, security review before release.	Backend/security reviewer
Third-party outage or latency	M/H	Timeouts, bounded retries, health monitoring, user-safe errors, queued/manual fallback, no hard dependency on optional AI.	Feature owner
AI output is inaccurate or unsuitable	M/M	Human review, editable/dismissible outputs, quality sampling, prompt/version records, rate/cost limits.	AI feature owner
Railway resource limitations, service interruptions, or delayed background-job execution	M/H	Monitor resource consumption, application health, and background-job execution; configure appropriate retry and recovery mechanisms; and maintain documented redeployment procedures.	Deployment Owner / Backend Developer
Migration or data-loss regression	M/H	Versioned Flyway migrations, representative tests, backup/export before risky changes, no destructive ad hoc production edits.	Backend/data reviewer
Cache exposes stale/cross-session data	M/H	User/role-aware query keys, mutation invalidation, logout clearing, regression tests, no sensitive persistent cache.	Frontend reviewer
Validation findings are not reflected in requirements/design	H/H	Maintain evidence-to-objective-to-requirement/design/test traceability; review before accepting implementation.	Project team / adviser
Insufficient test or deployment evidence	M/H	Complete the STD, retain execution/usability results, identify release baseline, and record unresolved conditions.	QA / deployment owner
Scope growth and late feedback	H/M	Change control, prioritize MVP acceptance, defer optional enhancements, update schedule and traceability.	Sprint coordinator
Insufficient validation participation	M/M	Recruit early, identify roles, schedule follow-ups, preserve evidence, report sample limitations.	Validation coordinator
Academic deadline / shared workload	H/M	Small PRs, visible blockers, shared review, milestone buffers, prioritize required deliverables.	Project team

5.3.8 Project Closeout Plan

- Confirm required deliverables and acceptance evidence are complete.
- Close or explicitly defer defects with documented impact and ownership.
- Tag/archive the accepted source and record environment/configuration requirements without secrets.
- Transfer operational guidance, Page authorization steps, backup/export notes, and fallback procedures.
- Revoke temporary access and tester credentials that are no longer required.
- Record lessons learned, remaining risks, and recommended post-capstone work.

6. TECHNICAL PROCESS PLANS

6.1 Process Model

The project uses iterative Agile development with Scrum-informed two-week cycles. Each increment follows: review validation evidence and approved requirements; clarify acceptance conditions; assess architecture, security, data, UI, and deployment impact; implement on a branch; run focused and regression tests; peer review through a pull request; merge into dev; verify in the integrated or deployed environment; collect stakeholder feedback; update the SRS, RTM, SDD, SPMP, STD, and priorities when affected.

The process is risk-adjusted. Authentication, authorization, tenant isolation, migrations, publishing state transitions, and external-token handling require broader test and review evidence than isolated presentation changes.

6.2 Methods, Tools, and Techniques

Category	Current method/tool
Requirements/design	Proposal, SRS, SDD, implementation plans, use cases, diagrams, traceability.
Frontend	React 19, TypeScript 6, Vite 8, React Router 7, TanStack Query 5, Axios, FullCalendar, Tailwind CSS.
Backend	Java 21, Spring Boot 4.0.6, Spring MVC/Security/Validation/Data JPA/Cache.
Data/storage	PostgreSQL/Supabase, Flyway, pgvector-dependent retrieval, Cloudflare R2.
External services	Facebook Graph API v25.0, Gemini, Voyage AI, Resend, Messenger integration.
Testing/quality	Spring tests, H2 test support, Playwright, ESLint, TypeScript build, bundle budgets, manual role workflows.
Configuration/deployment	Git/GitHub pull request workflow, Vercel for frontend deployment, Railway for backend deployment, environment-managed secrets, and application health monitoring.

6.3 Infrastructure Plan

The frontend is built and deployed independently from the Spring Boot REST backend. PostgreSQL stores transactional and audit data; R2 stores media objects; external APIs provide Page publishing, email, AI, embeddings, and messaging. Development, test, and deployment configuration is separated through environment variables. Secrets are never committed or copied into documentation.

Operational controls include health endpoints, background-job records, audit logs, bounded connection resources, retention jobs, publishing recovery paths, and performance regression checks. Backup, export, credential rotation, and production support ownership must be confirmed during transition.

6.4 Product Acceptance Plan

Acceptance area	Required evidence
Functional	Role-based primary workflows complete; state transitions and fallbacks behave as approved.
Security/privacy	Authentication, authorization, tenant boundaries, token/secret handling, auditability, and logout/session behavior verified.
Reliability	Publishing/email/AI failures degrade safely; retries are bounded; manual workflow remains available.
Performance	Critical SRS targets and frontend regression checks are measured; no invented results are reported.
Usability	Role-based usability tasks completed, task success rates and user feedback analyzed, and System Usability Scale (SUS) results evaluated against the minimum target score of 70 out of 100.
Validation traceability	Findings and approved priorities trace through SMART objectives, SRS, SDD, implementation, tests, and acceptance.
Documentation	SRS, RTM, SDD, SPMP, STD, validation artifacts, deployment guidance, and known limitations align with the accepted build.
Stakeholder	Required academic and operational reviewers record acceptance or actionable conditions.

7. SUPPORTING PROCESS PLANS

7.1 Configuration Management Plan

Git is the configuration-management system. Work begins from the latest dev baseline on a named branch. Commits remain scoped, pull requests describe behavior and files, checks are completed, and reviewed work is merged. Database migrations are immutable once shared. Release baselines identify the source commit, migration state, deployed environments, and matching document versions.

7.2 Verification and Validation Plan

Verification checks that requirements are represented in design, code, migrations, tests, and documentation. Validation checks that intended users can complete representative tasks and that measured outcomes address the project objectives. Testing covers unit/service/controller behavior, integration boundaries, frontend end-to-end paths, role/tenant access, failure paths, slow-network behavior where applicable, and regression checks after refactoring.

The five MVP Validation deliverables are complete. Their findings and evidence are the required input to the Refactoring Priorities, Updated SRS and RTM, Updated SDD, Updated SPMP, full implementation, deployment, and testing. The Week 9 Software Test Documents consist of the Test Plan, Test Cases, and Test Reports. The Week 10 deployed-system evidence includes test execution and usability results.

Stage	Verification/validation activity	Output
Requirements	Review clarity, consistency, traceability, and measurable acceptance.	Approved requirement/change record
Design	Review architecture, data, interfaces, security, failure handling, and deployability.	SDD/decision update
Implementation	Automated tests, lint/build, peer review, migration checks, focused manual scenarios.	PR evidence and test results
Integrated system	Role workflows, Page/AI/email/storage boundaries, recovery, performance, and accessibility checks.	System test evidence and defect log
MVP/UAT	Assigned role tasks, task success/time, SUS, AI usefulness/quality, and open feedback.	Responses, highlights, recommendations
Acceptance	Trace requirements and critical defects to final evidence.	Acceptance record and release baseline

7.3 Documentation Plan

Document	Status	Control
Proposal / objectives	Existing baseline	Update only through approved academic revision.
MVP Validation package	COMPLETED	Preserve original form, framework, responses, highlights, and evidence.
Refactoring Priorities	IN PROGRESS	Each priority must cite validation evidence and an affected objective/metric.
Updated SRS and RTM	COMPLETED	Requirements source of record; trace findings, objectives, features, design, and tests.
Updated SDD	COMPLETED	Align architecture, data, interfaces, workflows, security, and integrations with the accepted SRS.
Updated SPMP	COMPLETED	Update scope, schedule, WBS, resources, roles, risks, validation, and transition.
Software Test Documents	IN PROGRESS	Include Test Plan, Test Cases, Test Reports, requirement links, and measured results.
Deployment / user guidance	IN PROGRESS / UPCOMING	Match the deployed build, include operational guidance, and exclude secrets.

7.4 Quality Assurance Plan

Quality assurance combines preventive standards and evidence-based review: scoped changes, established repository patterns, authorization at the server, safe migrations, clear error/loading behavior, tests proportional to risk, review before merge, and documentation alignment. The team does not accept a passing build as proof of functional, security, or usability acceptance.

7.5 Reviews and Audits

- Peer code review for each pull request.

- Requirements/design review when behavior or architecture changes.
- Security and tenant-scope review for identity, role, institution, token, and media access changes.
- Migration review for schema/data changes.
- Performance and cache review for route data, refetch, invalidation, and cross-session behavior.
- Milestone audit of deliverables, evidence, known limitations, and unresolved risks.

7.6 Problem Resolution Plan

Problems are recorded with reproducible steps, affected role/data, severity, environment, expected and observed behavior, evidence, and owner. Critical security, data-loss, cross-tenant, or publication-integrity issues receive immediate containment. Corrections include regression tests where practical and are verified in the integrated environment before closure.

7.7 Subcontractor Management Plan

The project has no software-development subcontractor. Cloud and API providers are managed as external dependencies through documented configuration, least-privilege access, service limits, privacy review, fallback behavior, and monitoring. Provider terms do not transfer project acceptance responsibility away from the team.

7.8 Process Improvement Plan

At sprint and milestone reviews, the team identifies recurring defects, review delays, flaky tests, unclear requirements, deployment failures, cache regressions, and documentation drift. Improvements are converted into specific tasks, assigned, verified, and retained when they reduce rework without weakening controls.

8. ADDITIONAL PLANS

8.1 Security, Privacy, and Data Governance

- Apply least privilege and server-side role/tenant enforcement; never trust client-supplied institution identity.
- Protect passwords, JWT secrets, Meta tokens, AI keys, storage credentials, and email credentials; never place secret values in documents or Git.
- Maintain auditable state changes and privileged cross-institution activity.
- Use controlled retention and deletion behavior for user, media, audit, token, and job-run data.
- Use representative or consented test data and minimize personal information in AI prompts and evidence.
- Clear or invalidate user-specific cached data at logout/account/role/institution changes.

8.2 Deployment and Transition

Development deployment uses Vercel for the frontend and Railway for the backend, with Supabase PostgreSQL and Cloudflare R2. Production transition requires an accepted release baseline, environment inventory, migration verification, secret provisioning, health checks, backup/export confirmation, Meta Page authorization, monitoring ownership, and a rollback or manual fallback procedure.

The course milestone requires the system to be deployed and ready for testing by 17 October 2026, supported by test execution and usability test results. Functional, integration, system, and acceptance testing, together with adviser and user feedback, must be retained as release evidence. Deployment completion is not claimed until the URL, environment, source baseline, migration state, availability, and supporting results are confirmed.

Meta live operation is an organizational dependency. The DASIG Page administrator or authorized business representative must grant suitable access and complete any required Meta verification/review. Until that occurs, testing must use approved development/test identities and a test Page, and public production publishing must not be represented as complete.

Following the academic deployment milestone, the team plans to conduct an extended pilot deployment beginning in late October 2026 and continuing through November 2026, with operational activities potentially extending into December depending on testing requirements, stakeholder feedback, and resource availability. This period will support functional verification, usability evaluation, defect resolution, performance monitoring, and preparation for final acceptance and transition. The deployment schedule remains subject to academic requirements and stakeholder approval.

8.3 Subscription and AI Credit Plan

DASIGConnect plans to introduce institution-based subscription tiers to support the system's continuing operational, hosting, storage, maintenance, and AI-service expenses. The proposed plans are subject to adjustment after evaluating actual usage during pilot deployment.

Feature	Free	Plus	Premium
Proposed Monthly Fee	0	499	999
Users per institution	3	10	30
Monthly Submissions	50	300	1000
Media Storage	1 GB	10 GB	50 GB
AI caption credits per day	5	25	75
AI caption credits per month	100	600	2000
AI-processed media per month	100	100	5000
Analytics retention	30 days	12 months	Full retention
Data export	Not included	CSV	CSV and PDF
Support	Standard	Priority Email	Priority Support

One AI caption credit represents one successful caption-generation request. Regenerating a caption consumes another credit, while failed or timed-out AI requests will not be counted. Daily credits will reset at midnight and will not carry over to the following day.

Essential functions, including submission creation, review and approval, scheduling, notifications, role-based access, and account security, will remain available under all plans. Paid subscriptions will primarily provide greater capacity, AI usage, storage, analytics retention, exports, and support.

The proposed subscription model will initially undergo pilot evaluation. Pricing, limits, and inclusions will be reviewed based on actual institutional activity, AI consumption, infrastructure expenses, and stakeholder feedback. Implementation of subscription billing and enforcement is considered a future development and is not part of the currently completed MVP.

9. PLAN ANNEXES

9.1 Refactoring Impact Summary

Area	29 May Module 3 baseline	4 October refactored baseline
Scale	Commit ce776d2.	Substantial multi-module refactor and hardening through commit 3976afb; use Git/PR evidence rather than effort estimates from commit counts.
Roles	contributor / validator / administrator	contributor / moderator / admin; migrations repair and rename legacy values; moderator is network-wide.
Account governance	Initial authentication/provisioning scope.	Expanded password/session security, invitation provenance, role changes, admin promotion/transfer, and account controls.
Review	Validation queue and terminal decisions.	Editorial edits, self-review prevention, edit severity/history, missed-review handling, and lock cleanup.

Area	29 May Module 3 baseline	4 October refactored baseline
Media	Central repository and recommendation basis.	R2 storage, albums/nesting, staging, content hashes, display titles, history, retention, proxying, and watermark settings.
Publishing/engagement	Calendar, publishing, fallback, and exception handling.	OAuth/token validation, suspension/escalation, failed-publication recovery, engagement sync, and Page analytics.
Operations	Basic deployment and workflow analytics.	Audit query surfaces, append-only controls, system health, scheduled-job records, retention, and performance regression checks.
Frontend data behavior	Primarily page/component fetching.	TanStack Query caching, invalidation, route data reuse, and targeted performance tests.

9.2 Requirements-to-Management Traceability

Requirement/objective	Managed through	Evidence
Multi-institution role access and isolation	Scope, staffing, security, V&V, and risk plans.	Role/tenant tests, audit records, migration/source review.
Complete structured submissions	Content WBS, quality controls, metrics.	Validation behavior and completeness metric.
Mandatory review before publishing	Review/publishing WBS, acceptance, risk controls.	State transition and publication audit evidence.
Scheduling and publish timeliness	Integration plan, metrics, external-dependency risk.	Calendar/guard-rail tests and publication attempts.
Media retrieval and reuse	Media WBS, infrastructure, performance and lifecycle controls.	Search measurements and workflow tests.
AI assistance	AI governance, metrics, provider risk, optional-service fallback.	Interaction logs and validation ratings.
Usability and stakeholder fit	MVP/UAT plan, participant resources, validation deliverables.	Task results, SUS, comments, interviews, and recommendations.
Maintainability and live readiness	Configuration, documentation, QA, monitoring, transition.	Build/test checks, health/audit data, deployment and handover records.

9.3 Original May 2026 Module 3 Use-Case History

The following identifiers describe the presentation baseline merged to main on 29 May 2026. They are retained as historical evidence and are not relabeled using the later SRS numbering.

Original use case	Verified period	Principal contributors	Baseline output
UC-1.1 Authentication and Provisioning	19-23 May	Lerah, Mark, Chris, Richemmae	Authentication, provisioning, invitations, tests, route guards
UC-1.2 Institution Onboarding	19-29 May	Richemmae, Mark	Institution foundation, onboarding, scheduling, lifecycle, deletion
UC-1.3 Content Submission	20-29 May	Mark, Chris, Lerah, Richemmae	Composer, submission API, tags, guardrails, media, drafts, scheduling
UC-2.1 Content Validation	24-29 May	Lerah, Mark, Richemmae, Jay Lord	Locks, review actions, queue UI, revision handling
UC-2.2 Media Repository	23-29 May	Richemmae, Jay Lord, Chris	Upload, list/detail/delete, signed URLs, selection, repository UI
UC-2.3 Notifications	23-29 May	Richemmae, Chris, Jay Lord	Notification backend, SSE, filters, real-time UI
UC-2.4 Analytics Dashboard	27-29 May	Lerah, Jay Lord	Metrics APIs and role-scoped dashboard
UC-3.1 Calendar and Publishing	26-29 May	Chris, Richemmae	Calendar, Facebook publishing, scheduling, drag/drop
UC-3.2 AI Caption	26-29 May	Chris, Mark	Claude Vision caption generation and composer integration
UC-3.3 AI Media Recommendations	28-29 May	Chris	Recommendation service and media-picker integration
UC-3.4 Manual Publishing Fallback	27-29 May	Richemmae, Chris	Manual workflow, retry/recovery, resolution integration
UC-3.5 Administrator Exception Handling	27-29 May	Richemmae, Chris	Timeout, override, direct-post, token, exception workflows

9.4 Current SRS Use-Case Responsibility Matrix

Primary implementer means the strongest feature-specific Git evidence. Shared and follow-up work remains visible; the team is collectively accountable for review, testing, documentation, and acceptance.

Current SRS use case	Primary/shared implementers	Principal evidence
UC-1.1 Administrator Account Management	Chris; Richemmae follow-up	PRs 74, 104, 374
UC-1.2 Institution Management	Mark; Richemmae foundation/follow-up	PRs 78, 189, 191
UC-1.3 Contributor Account Management	Mark; Lerah and Chris foundations	PRs 76, 81
UC-1.4 Login and Session Management	Jay Lord; Lerah, Chris, Richemmae follow-up	PR 79 and later security fixes
UC-1.5 Post Drafting and Templates	Mark, Chris, Lerah; Richemmae follow-up	PRs 21, 23, 85
UC-1.6 AI and Text Tools Integration	Chris; Mark follow-up	PR 103
UC-1.7 Media Attachment Rules	Jay Lord; Richemmae and Chris follow-up	PR 80; PRs 203, 333-373
UC-1.8 Engagement Helpers	Jay Lord	PR 84
UC-1.9 Draft Submission	Lerah; Richemmae and Chris follow-up	PRs 77, 85
UC-1.10 Moderator Account Management	Chris and Richemmae	PRs 130, 135, 374
UC-2.1 Library Uploads and Albums	Lerah; Jay Lord foundation; Richemmae/Chris follow-up	PR 112 and media lifecycle work
UC-2.2 Semantic Search and Filtering	Jay Lord, Lerah, Chris, Richemmae	PRs 112, 310, 365-373
UC-2.3 Media History Tracking	Richemmae	PR 221
UC-2.4 Approval Workflow	Lerah, Mark, Richemmae; Chris performance follow-up	PRs 40, 42, 113
UC-2.5 Automated Watermarking	Mark; Richemmae/Chris follow-up	PRs 98, 267
UC-3.1 Master Calendar Visibility	Mark; Richemmae/Chris follow-up	PR 100 commits and later calendar work
UC-3.2 Automated Facebook Publishing	Lerah; Richemmae follow-up	PR 99
UC-3.3 Automated Reminders and Alerts	Jay Lord; Chris/Richemmae foundation	PR 111
UC-3.4 Social Engagement Analytics	Richemmae; Chris follow-up	PR 101 and later analytics work
UC-3.5 System Health Analytics	Chris; Richemmae follow-up	PRs 114, 118
UC-3.6 Audit Log Review	Jay Lord; Richemmae/Chris follow-up	PR 127 and later controls

9.5 Cross-Cutting Development Work

Work package	Principal contributor evidence	Management significance
Repository, deployment, CI, and database foundation	Richemmae; Chris support	Established source, environments, migrations, CI, and testing baseline.
R2 media-storage migration	Richemmae, PR 129	Changed upload, media lifecycle, publishing, and tenant-scoped storage.
Contributor/Moderator/Admin role refactor	Chris PR 130; Richemmae follow-up	Changed authorization, terminology, account management, and review responsibility.
Frontend caching architecture	Chris, PRs 164-187	Introduced session-aware caching, invalidation, and route data reuse.
React/backend performance program	Chris, PRs 213-258 and 312-327	Added cancellation, pagination, progressive loading, telemetry, and regression checks.
Responsive and interaction quality	Mark, May-September PRs	Improved cross-role mobile/desktop usability without changing core authorization.
Media AI and semantic retrieval	Chris, PRs 333-373	Added durable processing, multi-image activation, Voyage retrieval/ranking, evaluation, and monitoring.
Requirements reconciliation	Richemmae, September documentation PRs	Aligned living documents with current roles and implemented behavior.

9.6 Current Management Assessment

The DASIGConnect project has completed the five required MVP Validation outputs, along with the updated Software Requirements Specification (SRS), Software Design Description (SDD), and Software Project Management Plan (SPMP). These documents provide the foundation for aligning the system's implementation with approved requirements, stakeholder feedback, and project objectives.

The remaining project activities focus on completing validation-driven refactoring priorities, Software Test Documents, system testing, deployment preparation, and usability evaluation. The team will continue verifying implemented features against the approved requirements, documenting test results, addressing identified defects, and ensuring that all necessary evidence is available for final project acceptance and transition. Any remaining deployment dependencies, academic requirements, and stakeholder approvals will be monitored and addressed before project completion.

10. INDEX

Key topics are discoverable through the Table of Contents and the Definitions, Risk Management, Refactoring Impact, and Requirements-to-Management Traceability sections.